

EXPERIMENT 29: PROLOG PROGRAM FOR MEDICAL DIAGNOSIS

Aim

To write and execute a Prolog program for medical diagnosis using symptoms and rules.

Objective

- To represent medical knowledge using facts and rules.
- To identify diseases based on symptoms.
- To understand rule-based reasoning in Prolog.
- To develop a simple expert-system-like application.

Algorithm

1. Define symptoms as facts.
2. Define diseases and their associated conditions as rules.
3. Enter the required disease query.
4. Prolog checks whether the required symptoms are present.
5. If all conditions of a rule are satisfied, the disease is diagnosed.
6. Display the result.

Flowchart

START

|

v

Define Symptoms

|

v

Define Disease Rules

|

v

Enter Query

|

v

Check Symptoms

|

v

Conditions Satisfied?

/\

Yes No

| |

v v

Disease No

Found Diagnosis

|

v

STOP

Code

% Medical Diagnosis System

symptom(fever).

symptom(cough).

symptom(body_pain).

symptom(headache).

symptom(sneezing).

symptom(runny_nose).

symptom(stomach_pain).

symptom(vomiting).

disease(flu) :-

symptom(fever),

symptom(cough),

symptom(body_pain).

disease(common_cold) :-

 symptom(sneezing),

 symptom(runny_nose).

disease(migraine) :-

 symptom(headache).

disease(food_poisoning) :-

 symptom(stomach_pain),

 symptom(vomiting).

Query

?- disease(flu).

Output

true.

Another query:

?- disease(Disease).

Output

Disease = flu ;

Disease = common_cold ;

Disease = migraine ;

Disease = food_poisoning.

Result

The program successfully diagnoses diseases based on predefined symptoms and rules.

Conclusion

Thus, a simple medical diagnosis system was successfully implemented using Prolog.

```
1 % Medical Diagnosis System
2
3 symptom(fever).
4 symptom(cough).
5 symptom(body_pain).
6 symptom(headache).
7 symptom(sneezing).
8 symptom(runny_nose).
9 symptom(stomach_pain).
10 symptom(vomiting).
11
12 disease(flu) :-
13     symptom(fever),
14     symptom(cough),
15     symptom(body_pain).
16
17 disease(common_cold) :-
18     symptom(sneezing),
19     symptom(runny_nose).
20
21 disease(migraine) :-
22     symptom(headache).
23
24 disease(food_poisoning) :-
25     symptom(stomach_pain),
26     symptom(vomiting).
27
```

```
7 ?- make.
```

```
% c:/users/varshini m/onedrive/desktop/here!!!/  
true.
```

```
8 ?- disease(flu).
```

```
true.
```

```
9 ?- disease(Disease).
```

```
Disease = flu .
```

```
10 ?- disease(common_cold).
```

```
true.
```